iLearn Practice Exam

11BR · Chapter / Topic 5

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Topic 5: System Software

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 74

Marks obtained: ..... / 74

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 74.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 8 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A student has purchased a new laptop.

(a) The laptop has systems software.

(i) Describe how the Operating System (OS) manages processes in the computer.

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..... [5]

(ii) Describe the purpose of utility software in a computer.

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..... [2]

2 A software developer is working in a team writing a program for a client.

(a) The developer is writing a new program library to be used by the other team members.

(i) Define the term **program library**.

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..... [2]

(ii) Explain **two** benefits to the developer of choosing to create a Dynamic Link Library (DLL).

1 .....

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2 .....

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[4]

(b) The development team needs to use a translator whilst writing the program for the client.

Identify whether an interpreter or a compiler would be more appropriate at this stage of the program development.

Justify your choice.

Translator .....

Justification .....

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..... [3]

(c) The development team uses an Integrated Development Environment (IDE).

Complete the table by describing the typical features found in an IDE.

Feature	Description
Breakpoints	....
Dynamic syntax checks	....
Context-sensitive prompts	....
Single stepping	....

[4]

3 A computer has an Operating System (OS).

Memory management and process management are two OS tasks.

Explain how memory management **and** process management support multi-tasking.

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..... [4]

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4 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.

(a) The programmer decides to use the compiler when testing the final program.

Describe the benefits of using the compiler during testing.

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..... [2]

(b) IDEs have many features other than built-in translators.

Complete the table by identifying **one other** common IDE feature that can be used for each purpose. Describe how each feature helps the user during program development.

Each feature must be different. Do **not** give translator as one of your features.

Purpose	IDE feature	Description
for coding	....	....
for presentation	....	....
for debugging	....	....

[6]

(c) The programmer uses program libraries when developing the program.

Describe **two** benefits to the programmer of using program libraries.

1 .....

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2 .....

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[2]

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5 A student is writing a computer program for a school project.

(a) The student uses a computer that comes with an Operating System (OS) and utility software.

(i) Identify **two** tasks the OS performs as part of process management.

1 .....

2 .....
[2]

(ii) Identify **two** tasks the OS performs as part of memory management.

1 .....

2 .....
[2]

(iii) Explain the reasons why defragmentation software improves the performance of a computer.

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..... [3]

(b) The student is using an Integrated Development Environment (IDE).

(i) Explain how the student can use debugging tools in the IDE.

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..... [4]

(ii) The high-level language the student is using can be partially compiled and partially interpreted.

Explain the reasons why some high-level languages are partially compiled and partially interpreted.

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..... [2]

6 A student uses a laptop for school work. The laptop contains system software including an Operating System (OS) and utility software.

(a) Hardware management is one key management task that the OS performs.

(i) Identify **two** tasks the OS performs as part of hardware management.

1 .....

2 .....
[2]

(ii) Identify **two** other key management tasks that the OS performs.

1 .....

2 .....
[2]

(b) A disk formatter is an example of utility software.

(i) State the purpose of utility software in a computer.

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..... [1]

(ii) Describe the purpose of a disk formatter.

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..... [2]

(c) The student is writing a computer program in a high-level language.

The student imports routines from program libraries into their program.

(i) Define the term program library.

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..... [1]

(ii) Describe **one** benefit of the student using a Dynamic Link Library (DLL) instead of a non-Dynamic Link Library.

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..... [2]

7 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

(a) Describe the following features of a typical IDE.

Context-sensitive prompts .....

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Single stepping .....

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[4]

(b) The program is distributed by downloading the source code and its library files from a web server.

(i) Explain the reasons for compressing the files.

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..... [2]

(ii) The program files are stored on a new hard disk after they have been downloaded.

Describe the reasons why a hard disk formatter is needed for the new hard disk.

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..... [3]

8 Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(a) State **two** disadvantages of using a compiler compared to an interpreter during program development.

1 .....

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2 .....

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[2]

(b) Explain how a programmer benefits from using program libraries during program development.

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..... [3]

(c) Memory management is one key management task performed by an OS.

Give **three** other key management tasks that are performed by an OS.

1 .....

2 .....

3 .....

[3]

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END OF QUESTION PAPER